

Day 10 — Final Assessment + Project Showcase

Module: $\times 11$ (*Ekādhikena Pūrveṇa*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will demonstrate mastery of the $\times 11$ family of techniques via a summative quiz and a 2-minute project presentation.

Materials

- Summative quiz (`quizzes/summative.md`).
- Stopwatch.
- Rubric (`assessments/rubric.md`) per pair.
- Reflection slip per student.

Lesson flow

Warm-up (2 min)

- “Quiz first (15 min), then presentations.”

Part 1: Summative Quiz (15 min)

- Distribute `quizzes/summative.md`. Closed notebook.
- Covers: 2-digit $\times 11$ (with and without carry), 3-digit $\times 11$, *Anurūpyeṇa* extension, the algebraic identity, the historicity question (1 short-answer).
- Collect at 15-min mark.

Part 2: Project Presentations (25 min)

- 2-min presentation + 1-min Q&A per pair (3 min total per pair).
- ~8 pairs fit in 25 min. Reserve next class for remaining pairs if needed.
- During each presentation:
 - Teacher marks rubric in real-time.
 - One assigned audience member asks one question.
 - Other audience members fill “I learned / I wonder” slips.

Wrap-up (3 min)

- Closing question (oral): “What’s one mental-math habit you’ll keep using after this module?”
- Reflection slip submitted (reproduced in `student-workbook.md`).

Student-facing content

Today: quiz + presentations

15 min for the summative quiz (closed book).

2-minute presentation per pair, then 1-min audience question.

Reflection (due at end of class): 1. What was the most surprising thing you learned about $\times 11$? 2. Did your view of “Vedic mathematics” change during this module? How? 3. If you had three more days, what would you do differently?

Homework

- None. Module is complete.

Differentiation

- **Tier up:** Students finishing the quiz early can attempt a bonus “challenge” sheet — 4-digit $\times 11$ plus a 2-digit $\times 19$ mix.
- **Tier down:** Open-notebook quiz for students with documented learning needs.

Teacher notes

- Mark the quiz immediately after class while the presentations are fresh.
- Don’t rush the reflection — the “did your view change?” question is the cultural-honesty payoff of the module.
- If presentations spill into Day 11, protect that time.

Citation(s) used in this lesson

- All sources verified across the module.